Exercises from Ch. 6: Galois Theory, Lang

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Disclaimer: The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

Exercises

Exercise 0.1. (Exercise 1)
What is the Galois group of the following polynomials?

a) $X^3 - X - 1$ over $\mathbb{Q}$.

b) $X^3 - 10$ over $\mathbb{Q}$.

c) $X^3 - 10$ over $\mathbb{Q}(\sqrt{2})$.

d) $X^3 - 10$ over $\mathbb{Q}(\sqrt{-3})$.

e) $X^3 - X - 1$ over $\mathbb{Q}(\sqrt{-23})$.

f) $X^4 - 5$ over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{-5})$, $\mathbb{Q}(i)$.

g) $X^4 - a$ where a is any integer $\neq 0, \neq \pm 1$ and is square free. Over $\mathbb{Q}$.

h) $X^3 - a$ where a is any square-free integer ≥ 2. Over $\mathbb{Q}$.

i) $X^4 + 2$ over $\mathbb{Q}$, $\mathbb{Q}(i)$.

j) $(X^2 - 2)(X^2 - 3)(X^2 - 5)(X^2 - 7)$ over $\mathbb{Q}$.

k) Let $p_1, \dots, p_n$ be distinct prime numbers. What is the Galois group of $(X^2 - p_1) \cdots (X^2 - p_n)$ over $\mathbb{Q}$?

l) $(X^3 - 2)(X^3 - 3)(X^2 - 2)$ over $\mathbb{Q}(\sqrt{-3})$.

m) $X^n - t$, where t is transcendental over the complex numbers $\mathbb{C}$ and n is a positive integers. Over $\mathbb{C}(t)$.

n) $X^4 - t$, where t is as before. Over $\mathbb{R}(t)$.

Proof. a) By the Integral Root test, the polynomial $f(X) = X^3 - X - 1 \in \mathbb{Q}[X]$ is irreducible since ± 1 are not roots. The discriminant is $\Delta = -23$ which is not a square in $\mathbb{Q}$. Thus,

$$\mathrm{Gal}(f(X)/\mathbb{Q}) \cong S_3.$$

b) Let $f(X) = X^3 - 10 \in \mathbb{Q}[X]$. Since a cubic in characteristic $\neq 2, 3$ is irreducible if and only if there is no root in $\mathbb{Q}$, we use the Integral Root test. The possible rational roots of $f(X)$ are $\pm 1, \pm 2, \pm 5, \pm 10$, and none of these are roots of $f(X)$. Thus, $f(X)$ is irreducible over $\mathbb{Q}$. The discriminant is $\Delta = -2700$ which is not a square in $\mathbb{Q}$. Thus,

$$\mathrm{Gal}(f(X)/\mathbb{Q}) \cong S_3.$$

c) Again, since irreducibility of a cubic in a field of characteristic $\neq 2,3$ is dependent on if there is a root of $f(X)$ in that field, we suppose that $a+b\sqrt{2} \in \mathbb{Q}(\sqrt{2})$ is a root of $f(X)$. Therefore, $a+b\sqrt{2} = \sqrt[3]{10}$. In other words, suppose that $\sqrt[3]{10} \in \mathbb{Q}(\sqrt{2})$. Consider

$$\begin{aligned}\mathrm{Tr}_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}(\sqrt[3]{10}) &= \mathrm{Tr}_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}(a+b\sqrt{2}) \\ &= \sigma_1(a+b\sqrt{2}) + \sigma_2(a+b\sqrt{2}) \\ &= (a+b\sqrt{2}) + (a-b\sqrt{2}) \\ &= 2a \in \mathbb{Q}.\end{aligned}$$

The roots of $\mathrm{Irr}(\sqrt[3]{10}, \mathbb{Q}, X) = X^3 - 10$ are $\sqrt[3]{10}, \zeta_3\sqrt[3]{10}, \zeta_3^2\sqrt[3]{10}$. Thus,

$$\begin{aligned}\mathrm{Tr}_{\mathbb{Q}(\sqrt[3]{10})/\mathbb{Q}}(\sqrt[3]{10}) &= \tau_1(\sqrt[3]{10}) + \tau_2(\sqrt[3]{10}) + \tau_3(\sqrt[3]{10}) \\ &= \sqrt[3]{10} + \zeta_3\sqrt[3]{10} + \zeta_3^2\sqrt[3]{10} \\ &= \sqrt[3]{10}(1+\zeta+\zeta_3^2) = \sqrt[3]{10}\cdot 0 = 0\end{aligned}$$

Then by assumption we have the following tower $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{10}) \subseteq \mathbb{Q}(\sqrt{2})$, but our assumption also yields that $\mathbb{Q}(\sqrt[3]{10}) = \mathbb{Q}(\sqrt{2})$. Hence by the tower law for the trace,

$$\begin{aligned}\mathrm{Tr}_{\mathbb{Q}(\sqrt[3]{10})/\mathbb{Q}}(\sqrt[3]{10}) &= \mathrm{Tr}_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}\left(\mathrm{Tr}_{\mathbb{Q}(\sqrt[3]{10})/\mathbb{Q}(\sqrt{2})}(\sqrt[3]{10})\right) \\ 0 &= \mathrm{Tr}_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}(\sqrt[3]{10}) \\ &= \mathrm{Tr}_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}(a+b\sqrt{2}) = 2a.\end{aligned}$$

Thus, $2a = 0 \Rightarrow a = 0$. Hence we have that $a+b\sqrt{2} = b\sqrt{2} = \sqrt[3]{10}$. Therefore $2b^3\sqrt{2} = 10 \in \mathbb{Q}$ only if $b = 0$. But then $0 = 10$, an absurdity. Thus, $\sqrt[3]{10} \notin \mathbb{Q}(\sqrt{2})$, and $f(X) = X^3 - 10$ is irreducible over $\mathbb{Q}(\sqrt{2})$. The discriminant is $\Delta = -2700$ which is not a square in $\mathbb{Q}(\sqrt{2})$. Then

$$\mathrm{Gal}(f(X)/\mathbb{Q}(\sqrt{2})) \cong S_3.$$

d) We carry out a similar argument to that of part (c). Suppose that there exists some element $a+b\sqrt{-3} \in \mathbb{Q}(\sqrt{-3})$ such that $a+b\sqrt{-3} = \sqrt[3]{10}$, that is, $\sqrt[3]{10} \in \mathbb{Q}(\sqrt{-3})$. Then

$$\begin{aligned}\mathrm{Tr}_{\mathbb{Q}(\sqrt{-3})/\mathbb{Q}}(a+b\sqrt{-3}) &= \sigma_1(a+b\sqrt{-3}) + \sigma_2(a+b\sqrt{-3}) \\ &= (a+b\sqrt{-3}) + (a-b\sqrt{-3}) \\ &= 2a \in \mathbb{Q}.\end{aligned}$$

We also have that

$$\begin{aligned}\mathrm{Tr}_{\mathbb{Q}(\sqrt[3]{10})/\mathbb{Q}}(\sqrt[3]{10}) &= \tau_1(\sqrt[3]{10}) + \tau_2(\sqrt[3]{10}) + \tau_3(\sqrt[3]{10}) \\ &= \sqrt[3]{10} + \zeta_3\sqrt[3]{10} + \zeta_3^2\sqrt[3]{10} \\ &= \sqrt[3]{10}(1+\zeta_3+\zeta_3^2) = \sqrt[3]{10}\cdot 0 = 0.\end{aligned}$$

By the tower $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{10}) \subseteq \mathbb{Q}(\sqrt{-3})$ with equality in the last part of the tower, we have

$$\begin{aligned}\mathrm{Tr}_{\mathbb{Q}(\sqrt[3]{10})/\mathbb{Q}}(\sqrt[3]{10}) &= \mathrm{Tr}_{\mathbb{Q}(\sqrt{-3})/\mathbb{Q}}\left(\mathrm{Tr}_{\mathbb{Q}(\sqrt[3]{10})/\mathbb{Q}(\sqrt{-3})}(\sqrt[3]{10})\right) \\ 0 &= \mathrm{Tr}_{\mathbb{Q}(\sqrt{-3})/\mathbb{Q}}(\sqrt[3]{10}) \\ &= \mathrm{Tr}_{\mathbb{Q}(\sqrt{-3})/\mathbb{Q}}(a+b\sqrt{-3}) \\ &= 2a\end{aligned}$$

Thus, $2a = 0 \Rightarrow a = 0$. Then $a + b\sqrt{-3} = b\sqrt{-3} = \sqrt[3]{10}$ which yields that $-3b^3\sqrt{-3} = 10$ which is impossible since $-3b^3\sqrt{-3} \in \mathbb{C}$. Hence, $\sqrt[3]{10} \notin \mathbb{Q}(\sqrt{-3})$, so $X^3 - 10 \in \mathbb{Q}(\sqrt{-3})[X]$ is irreducible. The discriminant is $\Delta = -2700 = (30\sqrt{-3})^2$, which is a square in $\mathbb{Q}(\sqrt{-3})$. Therefore,

$$\operatorname{Gal}(X^3 - 10/\mathbb{Q}(\sqrt{-3})) \cong A_3.$$

Observe that $\sqrt{-3} \in 2\zeta_3 + 1$ and thus $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$. Then since the splitting field of $X^3 - 10$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[3]{10}, \zeta_3)$, we immediately see that

$$\begin{aligned} |\operatorname{Gal}(X^3 - 10/\mathbb{Q}(\sqrt{-3}))| &= |\operatorname{Gal}(\mathbb{Q}(\sqrt[3]{10}, \zeta_3)/\mathbb{Q}(\zeta_3))| \\ &= [\mathbb{Q}(\sqrt[3]{10}, \zeta_3) \colon \mathbb{Q}(\zeta_3)] = 3 \end{aligned}$$

which gives the isomorphism of the Galois group with $A_3 \cong \mathbb{Z}/3\mathbb{Z}$.

e) Let $f(X) = X^3 - X - 1$. Then $f(X)$ can be shown to be irreducible over $\mathbb{Q}(\sqrt{-23})$ (the discriminant is $\Delta = -23$) by assuming that some element $a + b\sqrt{-23} \in \mathbb{Q}(\sqrt{-23})$ is a root and reaching a contradiction. Thus we have that $f(X)$ is irreducible over $\mathbb{Q}(\sqrt{-23})$ with discriminant $\Delta = -23 = (\sqrt{-23})^2$ which is a square in $\mathbb{Q}(\sqrt{-23})$. Thus,

$$\operatorname{Gal}(f(X)/\mathbb{Q}(\sqrt{-23})) \cong A_3.$$

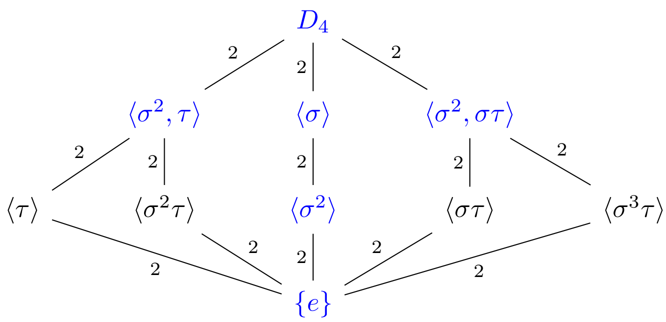
f) Set $f(X) = X^4 - 5$. Then the roots of $f(X)$ are $\pm\sqrt[4]{5}, \pm i\sqrt[4]{5}$. Hence the splitting field of $f(X)$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[4]{5}, i)$ and this is clearly normal and separable and therefore Galois over $\mathbb{Q}$. Since the elements of the Galois group permute the roots, there exists an automorphism such that

$$\sigma(x) = \begin{cases} i\sqrt[4]{5}, & x = \sqrt[4]{5} \\ i, & x = i \end{cases}$$

and there exists an automorphism τ such that

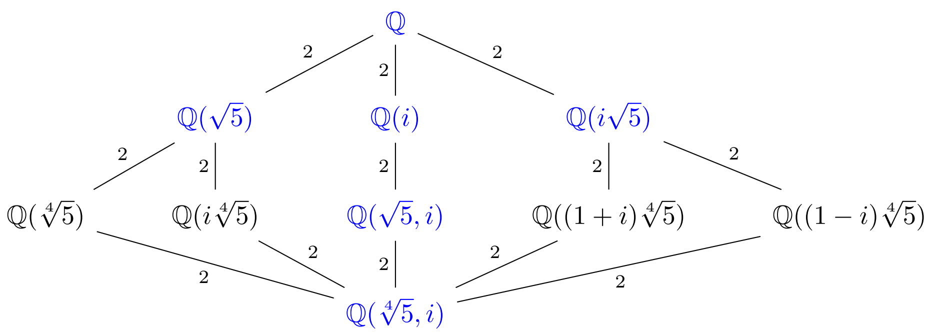
$$\tau(x) = \begin{cases} \sqrt[4]{5}, & x = \sqrt[4]{5} \\ -i, & x = i \end{cases}$$

Hence it is trivial to verify that $|\sigma| = 4$ and $|\tau| = 2$. Since $\langle\sigma\rangle \cap \langle\tau\rangle = \{e\}$, it follows at once that $\operatorname{Gal}(\mathbb{Q}(\sqrt[4]{5}, i)/\mathbb{Q}) \cong \langle\sigma, \tau\rangle \cong D_4$. Hence $\tau\sigma = \sigma^3\tau$ which is verifiable. By the Fundamental Theorem of Galois Theory, it will suffice to construct the subgroup lattice of the Galois group and then via the correspondence, construct the subfield lattice. The subgroup lattice of D_4 can be seen below:



The blue font denotes the normal subgroups of D_4 and the numbers are the indices of the lower subgroup in the upper subgroup. The first row below D_4 are all normal since they have index 2 while $\langle\sigma^2\rangle$ is normal since $Z(D_4) = \langle\sigma^2\rangle$.

We can now construct the subfield lattice of our Galois extension by the correspondence $H \mapsto \mathbb{Q}(\sqrt[4]{5}, i)^H$, the fixed field of H, for all $H \leqslant D_4$. Thus, we have the following subgroup lattice (inverted for ease of comparability with the subgroup lattice):



The blue in this lattice denotes the normal extensions of $\mathbb{Q}$ which correspond to the normal subgroups of D_4 and the numbers are the degrees of the lower field over the upper field. Note that in both lattices, we can compute indices (or degrees) for any subgroup (or subfield) via tower laws. The exercise now follows trivially from these lattices by the correspondence. Hence,

$$\begin{aligned}
\mathrm{Gal}(\mathbb{Q}(\sqrt[4]{5}, i)/\mathbb{Q}) &\cong D_4 \\
\mathrm{Gal}(\mathbb{Q}(\sqrt[4]{5}, i)/\mathbb{Q}(\sqrt{5})) &\cong \langle \sigma^2, \tau \rangle \cong V_4 \\
\mathrm{Gal}(\mathbb{Q}(\sqrt[4]{5}, i)/\mathbb{Q}(\sqrt{-5})) &\cong \langle \sigma^2, \sigma\tau \rangle \cong V_4 \\
\mathrm{Gal}(\mathbb{Q}(\sqrt[4]{5}, i)/\mathbb{Q}(i)) &\cong \langle \sigma \rangle \cong \mathbb{Z}/4\mathbb{Z}.
\end{aligned}$$

g) Let $f(X) = X^4 - a$ where a is any integer $\neq -, \pm 1$ and is square free. By part (f), we can generalize and see that the splitting field of $f(X)$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[4]{a}, i)$ which is a degree 8 extension with Galois group $\mathrm{Gal}(\mathbb{Q}(\sqrt[4]{a}, i)/\mathbb{Q}) \cong D_4$.

h) The cubic $X^3 - a$ with $a \geq 2$ a square free integer is irreducible. The discriminant is $\Delta = -27a^2$ which is not a square in $\mathbb{Q}$ and therefore $\mathrm{Gal}(X^3 - a/\mathbb{Q}) \cong S_3$. One could also observe that the splitting field for $X^3 - a$ over $\mathbb{Q}$ is $\mathbb{Q}(\zeta_3, \sqrt[3]{a})$ which is degree 6 with the automorphisms "cycling" the roots.

i) Let $f(X) = X^4 + 2 \in \mathbb{Q}[X]$. Observe that this is precisely part (g) with $a = -2$ which is square free. Hence

$$\mathrm{Gal}(f(X)/\mathbb{Q}) \cong D_4$$

and via the correspondence in part (g) and the lattices,

$$\mathrm{Gal}(f(X)/\mathbb{Q}(i)) \cong \langle \sigma \rangle \cong \mathbb{Z}/4\mathbb{Z}.$$

j) Let $f(X) = (X^2 - 2)(X^2 - 3)(X^2 - 5)(X^2 - 7) \in \mathbb{Q}[X]$. From left to right, label these quadratics as f_1, f_2, f_3, and f_4. The splitting fields for each of these quadratics over $\mathbb{Q}$ are:

$$\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{7}),$$

respectively. Observe that each of these quadratic extensions are Galois with all of them having Galois group isomorphic to $\mathbb{Z}/2\mathbb{Z}$. These splitting fields also intersect trivially and therefore by Corollary 1.15,

$$\mathrm{Gal}(\mathbb{Q}(\sqrt{2})\mathbb{Q}(\sqrt{3})\mathbb{Q}(\sqrt{5})\mathbb{Q}(\sqrt{7})/\mathbb{Q}) = \mathrm{Gal}(\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7})/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^4$$

k) This is a generalization of part (j) and therefore we can apply Corollary 1.15 and obtain

$$\mathrm{Gal}(\mathbb{Q}(\sqrt{p_1}, \ldots, \sqrt{p_n})/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^n.$$

l) ???

m) ??

n) ??

□

Exercise 0.2. (Exercise 2)
Find the Galois groups over $\mathbb{Q}$ of the following properties.

(a) X^3+X+1
(b) X^3-X+1
(g) X^3+X^2-2X-1
(c) X^3+2X+1
(d) X^3-2X+1
(e) X^3-X-1
(f) $X^3-12X+8$

Proof. We use the fact that irreducibility of a cubic over a field of characteristic $\neq 2, 3$ is equivalent to not having a root in that field.

a) The cubic X^3+X+1 is irreducible by the Integral Root Test, since the only possible rational roots are ± 1 which are not roots of X^3+X+1. Thus, $X^3+X+1 \in \mathbb{Q}[X]$ is irreducible. The discriminant is

$$\Delta = -4a^3 - 27b^2 = -4(1)^3 - 27(1)^2 = -31$$

which is not a square in $\mathbb{Q}$. Let $f(X) = X^3+X+1$. Thus,

$$\mathrm{Gal}(f(X)/\mathbb{Q}) \cong S_3.$$

b) Similar to part (a), by the Integral Root Test, the only possible rational roots are ± 1 which are not roots of X^3-X+1. Thus, X^3-X+1 is irreducible over $\mathbb{Q}$. The discriminant is

$$\Delta = -4a^3 - 27b^2 = -4(-1)^3 - 27(1)^2 = -23$$

which is not a square in $\mathbb{Q}$. Therefore if we set $f(X) = X^3-X+1 \in \mathbb{Q}[X]$, then

$$\mathrm{Gal}(f(X)/\mathbb{Q}) \cong S_3.$$

c) By the Integral Root Test, the only possible rational roots are ± 1 which are not roots of X^3+2X+1. Thus, X^3+2X+1 is irreducible. The discriminant is

$$\Delta = -4a^3 - 27b^2 = -4(2)^3 - 27(1)^2 = -59$$

which is not a square in $\mathbb{Q}$. Then let $f(X) = X^3+2X+1 \in \mathbb{Q}[X]$ which means that

$$\mathrm{Gal}(f(X)/\mathbb{Q}) \cong S_3.$$

d) Let $f(X) = X^3-2X+1 \in \mathbb{Q}[X]$. Observe that 1 is a root of $f(X)$ and therefore we have $f(X) = X^3-2X+1 = (X-1)(X^2+X-1)$. By the Integral Root Test, X^2+X-1 is irreducible and has discriminant 5. The splitting field is $\mathbb{Q}(\sqrt{5})$. Thus,

$$\mathrm{Gal}(f(X)/\mathbb{Q}) = \mathrm{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \cong C_2.$$

e) By the Integral Root Test, ± 1 are the possible rational roots of $f(X) = X^3-X-1$. Since neither of these are roots of $f(X)$, we have that $f(X)$ is irreducible over $\mathbb{Q}$. The discriminant of $f(X)$ is

$$\Delta = -4a^3 - 27b^2 = -4(-1)^3 - 27(-1)^2 = -23$$

which is not a square in $\mathbb{Q}$. Thus,

$$\mathrm{Gal}(f(X)/\mathbb{Q}) \cong S_3.$$

f) By the Integral Root Test, the possible rational roots are $\pm 1, \pm 2, \pm 4, \pm 8$ and none of these are roots of $f(X) = X^3 - 12X + 8 \in \mathbb{Q}[X]$. Then $f(X)$ is irreducible. The discriminant is

$$\Delta = -4a^3 - 27b^2 = -4(-12)^3 - 27(8)^2 = 5184$$

which is a square in $\mathbb{Q}$. Thus,

$$\mathrm{Gal}(f(X)/\mathbb{Q}) \cong A_3.$$

g) Let $f(X) = X^3 + X^2 - 2X - 1 \in \mathbb{Q}[X]$. Then $f(X0$ is irreducible over $\mathbb{Q}$ since it has no root in $\mathbb{Q}$. We complete the cube, to get a depressed cubic, by letting $X = Y - \frac{1}{3}$. Then $f(Y) = Y^3 - \frac{7}{3}Y - \frac{7}{27}$. The discriminant is

$$\Delta = -4a^3 - 27b^2 = -4\left(-\frac{7}{3}\right)^3 - 27\left(-\frac{7}{27}\right)^2 = 49$$

which is a square in $\mathbb{Q}$. Thus,

$$\mathrm{Gal}(f(X)/\mathbb{Q}) \cong A_3.$$

□

Exercise 0.3. (Exercise 3)
Let $k = \mathbb{C}(t)$ be the field of rational functions in one variable. Find the Galois group over k of the following polynomials:

(a) $X^3 + X + t$
(b) $X^3 - X + t$
(c) $X^3 + tX + 1$
(d) $X^3 - 2tX + t$
(e) $X^3 - X - t$
(f) $X^3 + t^2X - t^3$

Proof. We use the fact that irreducibility over $\mathbb{C}(t)$ is the same as irreducibility over $\mathbb{C}[t]$ by Gauss's Lemma.

a) Since $X^3 + X + t$ is linear in t, it is certainly irreducible. The discriminant is

$$\Delta = -4a^3 - 27b^2 = -4(1)^3 - 27(t)^2 = -4 - 27t^2$$

which is not a square in $\mathbb{C}(t)$. Thus, if $f(X) = X^3 + X + t$, we have

$$\mathrm{Gal}(f(X)/\mathbb{C}(t)) \cong S_3.$$

b) By the same reasoning of part (a), we see that $X^3 - X + t$ is linear in t and is therefore irreducible. The discriminant

$$\Delta = -4(-1)^3 - 27(t)^2 = 4 - 27t^2$$

which is not a square in $\mathbb{C}(t)$. Then let $f(X) = X^3 - X + t$. Hence

$$\mathrm{Gal}(f(X)/\mathbb{C}(t)) \cong S_3.$$

c) Let $f(X) = X^3 + tX + 1 \in \mathbb{C}(t)$. Then $f(X)$ is linear in t and hence irreducible. The discriminant is

$$\Delta = -4a^3 - 27b^2 = -4(t)^3 - 27(1)^2 = -4t^3 - 27$$

which is not a square. Thus,

$$\mathrm{Gal}(f(X)/\mathbb{C}(t)) \cong S_3.$$

d) Again, $f(X) = X^3 - 2tX + t \in \mathbb{C}(t)$ is irreducible since it is linear in t. The discriminant is

$$\Delta = -4a^3 - 27b^2 = -4(-2t)^3 - 27(t)^2 = 32t^3 - 27t^2$$

which is not a square in $\mathbb{C}(t)$ so

$$\mathrm{Gal}(f(X)/\mathbb{C}(t)) \cong S_3.$$

e) We have $f(X) = X^3 - X - t \in \mathbb{C}(t)$ which is linear in t and hence irreducible. The discriminant of $f(X)$ is

$$\Delta = -4a^3 - 27b^2 = -4(-1)^3 - 27(-t)^2 = 4 - 27t^2$$

which is not a square. Thus,

$$\mathrm{Gal}(f(X)/\mathbb{C}(t)) \cong S_3.$$

f) By way of contradiction suppose that $f(X) = X^3 + t^2X - t^3 \in \mathbb{C}(t)$ is reducible. Then

$$f(X) = X^3 + t^2X - t^3 = (X - \alpha(t))(X^2 + \beta(t)X + \gamma(t)).$$

Hence, $\alpha(t)\gamma(t) = t^3$, $\alpha(t) - \beta(t) = 0$, and $\gamma(t) - \alpha(t)\beta(t) = t^2$. Thus, $\alpha(t) = t$, $\gamma(t) = t^2$, and $\beta(t) = 0$. But then this would imply that t is a root, which is a contradiction. Therefore, $f(X)$ is irreducible. The discriminant is

$$\Delta = -4a^3 - 27b^2 = -4(t^2)^3 - 27(-t^3)^2 = -4t^6 - 27t^6 = -31t^6 = (\sqrt{-31}t^3)^2$$

is a square since $\sqrt{-31} \in \mathbb{C}$. Then since the discriminant is a square,

$$\mathrm{Gal}(f(X)/\mathbb{C}(t)) \cong A_3.$$

$\square$

Exercise 0.4. (Exercise 4) Let k be a field of characteristic $\neq 2$. Let $c \in k$, $c \notin k^2$. let $F = k(\sqrt{c})$. Let $\alpha = a + b\sqrt{c}$ with $a, b \in k$ and not both $a, b = 0$. Let $E = F(\sqrt{\alpha})$. Prove that the following conditions are equivalent.

1. E is Galois over k.
2. $E = F(\sqrt{\alpha'})$, where $\alpha' = a - b\sqrt{c}$.
3. Either $\alpha\alpha' = a^2 - cb^2 \in k^2$ or $c\alpha\alpha' \in k^2$.

Show that when these conditions are satisfied, then E is cyclic over k of degree 4 if and only if $c\alpha\alpha' \in k^2$.

Proof. By assumption, $c \notin k^2$, and thus $[F\colon k] = 2$. Hence, the nontrivial automorphism of F/k is given by $\sqrt{c} \mapsto -\sqrt{c}$. Hence, $\alpha \mapsto \alpha'$ under this automorphism.
($1 \implies 2$) Suppose that E/k is Galois. Then E/k is normal and thus every embedding of E in some k^a over k induces an automorphism of E. Let $\sigma \in \mathrm{Aut}(F/k)$ be the nontrivial automorphism given by $\alpha \mapsto \alpha'$. Thus, σ extends to an automorphism of $E = F(\sqrt{\alpha})$. Hence,

$$\begin{aligned}\sigma(\alpha) =& \sigma\left((\sqrt{\alpha})^2\right) = \sigma(\sqrt{\alpha})^2 = \alpha' \\ \Rightarrow& \sigma(\sqrt{\alpha}) = \pm\sqrt{\alpha'}.\end{aligned}$$

Hence, $\sqrt{\alpha'} \in E$ and therefore $F(\sqrt{\alpha'}) \subseteq E$. Then since $F(\sqrt{\alpha})$ and $F(\sqrt{\alpha'})$ are quadratic extensions of F, they must be equal. Hence,

$$F(\sqrt{\alpha}) = E = F(\sqrt{\alpha'}).$$

($2 \implies 3$) We first prove the following proposition:

Proposition 0.1. If $d \in k$ is a square in $F = k(\sqrt{c})$, then either $d \in k^2$ or $cd \in k^2$.

Proof. Let $d \in k$ be a square in F. Thus, there exists an element $x + y\sqrt{c} \in F$ such that

$$d = (x + y\sqrt{c})^2 = x^2 + y^2 + 2xy\sqrt{c} \in k.$$

Thus, for this to be in k, we require that $2xy\sqrt{c} = 0$. Since $\operatorname{char} k \neq 2$, this implies that $xy = 0$.
(Case 1: $y = 0$) Then $d = x^2 \in k^2$.
(Case 2: $x = 0$) Then $d = (y\sqrt{c})^2 = y^2 c$. Multiplying by c, we obtain

$$cd = y^2 c^c = (yc^2) \in k^2.$$

□

We return to the proof of ($2 \implies 3$). Assume that $F(\sqrt{\alpha}) = F(\sqrt{\alpha'})$. Then $\sqrt{\alpha'} \in F(\sqrt{\alpha})$. Hence, there exists some $x + y\sqrt{\alpha} \in F(\sqrt{\alpha})$ such that

$$\sqrt{\alpha'} = x + y\sqrt{\alpha}.$$

Thus,

$$\alpha' = x^2 + y^2 + 2xy\sqrt{\alpha} \in F.$$

Hence, we require $2xy\sqrt{\alpha} = 0$. Since $\operatorname{char} k \neq 2$, this means that we must have $xy = 0$.
(Case 1: $y = 0$) Then $\sqrt{\alpha'} = x \in F$. Thus, $\alpha' = x^2 \in F^2$. Multiplying by $\alpha \in F$, we obtain $\alpha\alpha' = (x\sqrt{\alpha})^2 = \alpha x^2 \in F^2$. But

$$\alpha\alpha' = a^2 - cb^2 \in k.$$

Then, by our proposition, since $\alpha\alpha' \in F^2$, we have $\alpha\alpha' \in k^2$ or $c\alpha\alpha' \in k^2$ as desired.
(Case 2: $x = 0$) Then $\sqrt{\alpha'} = y\sqrt{\alpha}$. Thus,

$$\alpha' = \alpha y^2.$$

Hence,

$$\frac{\alpha'}{\alpha} = y^2 \in F^2 \quad (\because\ y \in F)$$
$$\frac{(\alpha')^2}{\alpha\alpha'} = \frac{(a - b\sqrt{c})^2}{a^2 - cb^2} = y^2 \in F^2.$$

Since $(\alpha')^2 \in F^2$ and the quotient $\frac{(\alpha')^2}{\alpha\alpha'} = y^2 \in F^2$, it follows that

$$\alpha\alpha' = a^2 - cb^2 \in F^2.$$

Hence, since $\alpha\alpha' \in k$ and $\alpha\alpha' \in F^2$, by the proposition, $\alpha\alpha' \in k^2$ or $c\alpha\alpha' \in k^2$ as desired.
($3 \implies 1$) Assume that $\alpha\alpha' \in k^2$ or $c\alpha\alpha' \in k^2$.
(Case 1: $\alpha\alpha' \in k^2$) Then there exists some $d \in k$ such that $d^2 = \alpha\alpha'$. Hence

$$\alpha' = \frac{\alpha\alpha'}{\alpha} = \frac{d^2}{\alpha} = \left(\frac{d}{\sqrt{\alpha}}\right)^2.$$

Thus, $\sqrt{\alpha'} = \pm\frac{d}{\sqrt{\alpha}}$. Since $d \in k \subseteq F \subseteq F(\sqrt{\alpha})$ and $\sqrt{\alpha} \in F(\sqrt{\alpha})$, we get $\pm\frac{d}{\sqrt{\alpha}} \in F(\sqrt{\alpha})$. Hence, $\sqrt{\alpha'} \in F(\sqrt{\alpha})$.
(Case 2: $c\alpha\alpha' \in k^2$) Then there exists some $d \in k$ such that $d^2 = c\alpha\alpha'$. Hence,

$$\left(\frac{d}{\sqrt{c}\sqrt{\alpha}}\right)^2 = \frac{d^2}{c\alpha} = \frac{c\alpha\alpha'}{c\alpha} = \alpha'.$$

Thus, $\sqrt{\alpha'} = \pm\frac{d}{\sqrt{c}\sqrt{\alpha}}$. From $d \in k \subseteq F \subseteq F(\sqrt{\alpha})$, $c \in F \subseteq F(\sqrt{\alpha})$, and $\sqrt{\alpha} \in F(\sqrt{\alpha})$, it follows that $\pm\frac{d}{\sqrt{c}\sqrt{\alpha}} \in F(\sqrt{\alpha})$. Thus, $\sqrt{\alpha'} \in F(\sqrt{\alpha})$.

Therefore, in either case, $\sqrt{\alpha'} \in F(\sqrt{\alpha})$. Thus, from the above, $\sigma(\sqrt{\alpha}) = \pm\sqrt{\alpha'} \in E = F(\sqrt{\alpha})$. This is an extension of the automorphism of F/k given by $\sqrt{c} \mapsto -\sqrt{c}$. Then the conjugate of $\sqrt{\alpha}$ over k is $\pm\sqrt{\alpha'}$ and both of them lie in E. Hence, $\mathrm{Irr}(\sqrt{\alpha}, k, X)$ splits in E. Thus, E/k is normal. The extension E/k is separable since $\operatorname{char} k \neq 2$ and the tower

$$k \subseteq F \subseteq E$$

is quadratic at each step, and thus the irreducible polynomials of each respective extension are quadratic. Thus, from the characteristic of $k \neq 2$, the derivative of the irreducible polynomials at each step is nonzero, which implies that they have distinct roots. Therefore, E/k is separable and thus Galois.
We proceed to prove the last part. We prove the forward direction by proving the contrapositive.
($\Longrightarrow$) We see from the tower

$$k \subseteq F \subseteq E,$$

that $[E\colon k] = [E\colon F] \cdot [F\colon k] = 2 \cdot 2 = 4$. Thus, $\mathrm{Gal}(E/k)$ has order 4 and, therefore, is either cyclic or not cyclic, i.e. isomorphic to $\mathbb{Z}/4\mathbb{Z}$ or isomorphic to $V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (V_4 is the Klein four group). If $\alpha\alpha' \in k^2$, then from (2), $\sqrt{\alpha}, \sqrt{\alpha'} \in E$. The basis for E/k is $\mathcal{B} = \{1, \sqrt{\alpha}, \sqrt{\alpha'}, \sqrt{\alpha\alpha'}\}$. From our assumption that $\alpha\alpha' \in k^2$, we have $\sqrt{\alpha\alpha'} \in k$ which means that it is fixed by any automorphism of E over k. Hence, we have the following automorphisms:

$$\begin{aligned}
&\tau_1 = \mathrm{id} \\
&\tau_2\colon \sqrt{\alpha} \mapsto -\sqrt{\alpha} \quad \text{and} \quad \sqrt{\alpha'} \mapsto \sqrt{\alpha'} \\
&\tau_3\colon \sqrt{\alpha'} \mapsto -\sqrt{\alpha'} \quad \text{and} \quad \sqrt{\alpha} \mapsto \sqrt{\alpha} \\
&\tau_4 = \tau_2\tau_3\colon \sqrt{\alpha} \mapsto -\sqrt{\alpha} \quad \text{and} \quad \sqrt{\alpha'} \mapsto -\sqrt{\alpha'}
\end{aligned}$$

with $\sqrt{\alpha\alpha'} \mapsto \sqrt{\alpha\alpha'}$ for all τ_i. Thus, observe that for all $\tau_i \in \mathrm{Gal}(E/k)$, $|\tau_i| = 2$. Thus, $\mathrm{Gal}(E/k) \cong V_4$ if $\alpha\alpha' \in k^2$.
($\Longleftarrow$) Suppose $c\alpha\alpha' \in k^2$. Thus, there exists some $d \in k$ such that

$$\sqrt{\alpha'} = \frac{d}{\sqrt{c}\sqrt{\alpha}} \in E$$

Define $\sigma\colon E \to E$ by

$$\sigma(\sqrt{c}) = -\sqrt{c},\ \sigma(\sqrt{\alpha}) = \sqrt{\alpha'} = \frac{d}{\sqrt{c}\sqrt{\alpha}}.$$

We explicitly show that $|\sigma| = 4$. Thus,
<u>(σ^2)</u>:

$$\begin{aligned}
\sigma^2(\sqrt{\alpha}) &= \sigma(\sqrt{\alpha'}) \\
&= \sigma\left(\frac{d}{\sqrt{c}\sqrt{\alpha}}\right) \\
&= -\frac{d}{\sqrt{c}\sqrt{\alpha}} = -\frac{d}{\sqrt{c}\left(\frac{d}{\sqrt{c}\sqrt{\alpha}}\right)} \\
&= -\sqrt{\alpha}.
\end{aligned}$$

<u>(σ^3)</u>:

$$\begin{aligned}
\sigma^3(\sqrt{\alpha}) &= \sigma(\sigma^2(\sqrt{\alpha})) \\
&= \sigma(-\sqrt{\alpha}) = -\sigma(\sqrt{\alpha}) \\
&= -\sqrt{\alpha'}.
\end{aligned}$$

<u>(σ^4)</u>

$$
\begin{aligned}
\sigma^4(\sqrt{\alpha}) &= \sigma(\sigma^3(\sqrt{\alpha})) \\
&= \sigma(-\sqrt{\alpha'}) = -\sigma(\sqrt{\alpha'}) \\
&= -(-\sqrt{\alpha}) = \sqrt{\alpha}.
\end{aligned}
$$

Hence, $\sigma \in \mathrm{Gal}(E/k)$ with $|\sigma| = 4 = |\mathrm{Gal}(E/k)| = [E\colon k]$. Thus, $\mathrm{Gal}(E/k) = \langle\sigma\rangle \cong C_4 \cong \mathbb{Z}/4\mathbb{Z}$. Hence, E/k is cyclic of degree 4.

□

Exercise 0.5. (Exercise 5)
Let k be a field of characteristic $\neq 2, 3$. Let $f(X), g(X) = X^2 - c$ be irreducible polynomials over k, of degree 3 and 2 respectively. Let D be the discriminant of f. Assume that

$$
[k(D^{1/2})\colon k] = 2 \quad \text{and} \quad k(D^{1/2}) \neq k(c^{1/2}).
$$

Let α be a root of f and β a root of g in an algebraic closure. Prove:

a) The splitting field of fg over k has degree 12.

b) Let $\gamma = \alpha + \beta$. Then $[k(\gamma)\colon k] = 6$.

Proof. a) **PROOF IDEA IN SECTION NOTES FOR SIMPLICITY OF $SL_2(F)/\pm 1$**

b) ??

□

Exercise 0.6. (Exercise 6)

a) Let K be cyclic over k of degree 4, and of characteristic $\neq 2$. Let $G_{K/k} = \langle\sigma\rangle$. Let E be the unique subfield of K of degree 2 over k. Since $[K\colon E] = 2$, there exists $\alpha \in K$ such that $\alpha^2 = \gamma \in E$ and $K = E(\alpha)$. Prove that there exists $z \in E$ such that

$$
z\sigma z = -1, \quad \sigma\alpha = z\alpha, \quad z^2 = \sigma\gamma/\gamma.
$$

b) Conversely, let E be a quadratic extension of k and let $G_{E/k} = \langle\tau\rangle$. Let $z \in E$ be an element such that $z\tau z = -1$. Prove that there exists $\gamma \in E$ such that $z^2 = \tau\gamma/\gamma$. Then $E = k(\gamma)$. Let $\alpha^2 = \gamma$, and let $K = k(\alpha)$. Show that K is Galois, cyclic of degree 4 over k. Let σ be an extension of τ to K. Show that σ is an automorphism of K which generates $G_{K/k}$, satisfying $\sigma^2\alpha = -\alpha$ and $\sigma\alpha = \pm z\alpha$. Replacing z by $-z$ originally if necessary, one can then have $\sigma\alpha = z\alpha$.

Proof. a) Assume that K/k is cyclic of degree 4, that is $\mathrm{Gal}(K/k) \cong \mathbb{Z}/4\mathbb{Z}$ with the Galois group generated by σ. Then since $k \subseteq E \subseteq K$ where K/E is quadratic over E, the elements σ^2 fixes E. Then for an element z to be in E, we must have that $\sigma^2 z = z$. Let $\zeta \in K$ be a fourth root of unity in k. We have $X^2 - \gamma = \mathrm{Irr}(\alpha, E, X)$ which is separable since $\operatorname{char} k \neq 2$ and thus $\pm\alpha$ are its roots. It follows that we must have $\sigma^2\alpha = -\alpha$ since γ is fixed by σ^2. Define $z = \sigma\alpha/\alpha$. Then

$$
\begin{aligned}
\sigma^2 z &= \frac{\sigma^2(\sigma\alpha)}{\sigma^2\alpha} \\
&= \frac{\sigma(\sigma^2\alpha)}{\sigma^2\alpha} \\
&= \frac{\sigma(-\alpha)}{-\alpha} = \frac{-\sigma\alpha}{-\alpha} = z.
\end{aligned}
$$

Hence $z \in E$. By clearing the denominator, we obtain $\sigma\alpha = z\alpha$. Next,

$$
z^2 = \frac{(\sigma\alpha)^2}{\alpha^2} = \frac{\sigma(\alpha^2)}{\alpha^2} = \frac{\sigma\gamma}{\gamma}.
$$

Finally,

$$z\sigma z = \left(\frac{\sigma\alpha}{\alpha}\right)\sigma\left(\frac{\sigma\alpha}{\alpha}\right) = \left(\frac{\sigma\alpha}{\alpha}\right)\left(\frac{\sigma^2\alpha}{\sigma\alpha}\right) = \frac{\sigma^2\alpha}{\alpha} = \frac{-\alpha}{\alpha} = -1$$

as desired. Therefore $z = \sigma\alpha/\alpha \in E$ is the element we seek.

b) ???

$\square$

Exercise 0.7. (Exercise 7)

a) Let $K = \mathbb{Q}(\sqrt{a})$ where $a \in \mathbb{Z}$, $a < 0$. Show that K cannot be embedded in a cyclic extension whose degree over $\mathbb{Q}$ is divisible by 4.

b) Let $f(X) = X^4 + 30X^2 + 45$. Let α be a root of F. Prove that $\mathbb{Q}(\alpha)$ is cyclic of degree 4 over $\mathbb{Q}$.

c) Let $f(X) = X^4 + 4X^2 + 2$. Prove that f is irreducible over $\mathbb{Q}$ and that the Galois group is cyclic.

Proof. a) Let $K = \mathbb{Q}(\sqrt{a})$ with $a \in \mathbb{Z}$, $a < 0$. Thus, K is a imaginary quadratic extension of $\mathbb{Q}$, so $K \not\subseteq \mathbb{R}$. More succinctly, complex conjugation acts nontrivially on K.

By way of contradiction, assume that there exists a cyclic extension $L/\mathbb{Q}$ such that $4 \mid [L\colon \mathbb{Q}] \Rightarrow [L\colon \mathbb{Q}] = 4m$ for some positive integer m with $\mathbb{Q}(\sqrt{a}) \hookrightarrow L$. Then

$$\mathrm{Gal}(L/\mathbb{Q}) \cong C_{4m} \cong \mathbb{Z}/4m\mathbb{Z}.$$

Let $\tau \in \mathrm{Gal}(L/\mathbb{Q})$ denote complex conjugation, that is, $|\tau| = 2$ and, by construction, $\tau\mid_K \neq \mathrm{id}$. Since $\mathrm{Gal}(L/\mathbb{Q}) = \langle\sigma\rangle$ for some generator σ of order $4m$, there exists a unique element of order 2, namely σ^{2m}. Thus $\tau = \sigma^{2m}$. There is a unique subgroup of index 2 (normal) in $\mathrm{Gal}(L/\mathbb{Q})$ given by $\langle\sigma^2\rangle$. Then by the Fundamental Theorem of Galois Theory, the subfield of corresponding to $\langle\sigma^2\rangle \lhd \mathrm{Gal}(L/\mathbb{Q})$ is given by $L^{\langle\sigma^2\rangle} = F = \{x \in L\colon \gamma(x) = x,\ \forall\gamma \in \langle\sigma^2\rangle\}$. And by the correspondence, this is a unique quadratic extension of $\mathbb{Q}$ (from the index). Since K is also quadratic, it follows that $K = F$. But since $\tau \in \langle\sigma^2\rangle$ and by construction $\tau(F) = F$, F is a real quadratic extension of $\mathbb{Q}$ which contradicts the fact that K is an imaginary quadratic extension of $\mathbb{Q}$.

Therefore, $K = \mathbb{Q}(\sqrt{a})$ cannot be embedded into a cyclic extension $L/\mathbb{Q}$ of degree divisible by 4.

b) Let $f(X) = X^4 + 30X^2 + 45 \in \mathbb{Q}[X]$. By Eisenstein's for $p = 5$, $f(X)$ is irreducible over $\mathbb{Q}$. Let $Y = X^2$. Then $f(Y) = Y^2 + 30Y + 45$. Using the quadratic formula and let α denote a root of $f(X)$ we find $\alpha = \sqrt{-15 + 6\sqrt{5}}$. By Exercise 4, with $c = 5$, $\beta = \alpha^2 = -15 + 6\sqrt{5}$, and $\beta' = (\alpha^2)' = -15 - 6\sqrt{5}$ we have

$$\begin{aligned} c\beta\beta' &= 5(-15 + 6\sqrt{5})(-15 - 6\sqrt{5}) \\ &= 15^2 \in \mathbb{Q}^2. \end{aligned}$$

Therefore, $\mathbb{Q}(\alpha)/\mathbb{Q}$ is cyclic of degree 4.

c) Let $f(X) = X^4 + 4X^2 + 2 \in \mathbb{Q}[X]$. By Eisenstein's with $p = 2$, $f(X)$ is irreducible over $\mathbb{Q}$. Let $Y = X^2$, so $f(Y) = Y^2 + 4Y + 2$. Applying the quadratic formula, we find that a root of $f(X)$ is $\alpha = \sqrt{-2 + \sqrt{2}}$. Thus by Exercise 4, with $c = 2$, $\beta = \alpha^2 = -2 + \sqrt{2}$, and $\beta' = (\alpha^2)' = -2 - \sqrt{2}$ we get

$$\begin{aligned} c\beta\beta' &= 2(-2 + \sqrt{2})(-2 - \sqrt{2}) \\ &= 2^2 \in \mathbb{Q}^2. \end{aligned}$$

Therefore $\mathbb{Q}(\alpha)/\mathbb{Q}$ is cyclic of degree 4.

$\square$

Exercise 0.8. (Exercise 8)
Let $f(X) = X^4 + aX^2 + b$ be an irreducible polynomial over $\mathbb{Q}$, with roots $\pm\alpha, \pm\beta$, and splitting field K.

a) Show that $\mathrm{Gal}(K/\mathbb{Q})$ is isomorphic to a subgroup of D_8 (the non-abelian group of order 8 other than the quaternion group), and thus is isomorphic to on of the following:

   (i) $\mathbb{Z}/4\mathbb{Z}$ (ii) $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (iii) D_8.

b) Show that the first case happens if and only if

$$\frac{\alpha}{\beta} - \frac{\beta}{\alpha} \in \mathbb{Q}.$$

   Case (ii) happens if and only if $\alpha\beta \in \mathbb{Q}$ or $\alpha^2 - \beta^2 \in \mathbb{Q}$. Case (iii) happens otherwise. (Actually, in (ii), the case $\alpha^2 - \beta^2 \in \mathbb{Q}$ cannot occur. It corresponds to a subgroup of $D_8 \subset S_4$ which is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, but is not transitive on $\{1, 2, 3, 4\}$).

c) Find the splitting field K in $\mathbb{C}$ of the polynomial

$$X^4 - 4X^2 - 1.$$

   Determine the Galois group of this splitting field over $\mathbb{Q}$, and describe fully the lattices of subfields and of subgroups of the Galois group.

Proof. **SEE NOTES FOR WORK THUS FAR!!!**

a) ??

b) ??

c) ??

□

Exercise 0.9. (Exercise 15)
Let K/k be a Galois extension, and F an intermediate field between k and K. Let H be the subgroup of $\mathrm{Gal}(K/k)$ mapping F into itself. Show that H is the normalizer of $\mathrm{Gal}(K/F)$ in $\mathrm{Gal}(K/k)$.

Proof. Assume that K/k is Galois with $G = \mathrm{Gal}(K/k)$ and let $k \subseteq F \subseteq K$. Assume that

$$H := \{\sigma \in G \colon \sigma(F) = F\} \leqslant G.$$

We denote the normalizer of $\mathrm{Gal}(K/F)$ in $G = \mathrm{Gal}(K/k)$ by

$$N = N_G(\mathrm{Gal}(K/F)) = \{\sigma \in G \colon \sigma \,\mathrm{Gal}(K/F)\sigma^{-1} = \mathrm{Gal}(K/F)\} \leqslant G.$$

Thus, we want to show that $H = N$.
($H \subseteq N$) Let $\sigma \in H$ and $\tau \in \mathrm{Gal}(K/F)$. Then for all $\alpha \in F$, $\sigma^{-1}(\alpha) \in F$ and thus

$$\tau(\sigma^{-1}(\alpha)) = \sigma^{-1}(\alpha)$$

since $\mathrm{Gal}(K/F)$ fixes F. Applying σ on the left gives

$$\left(\sigma\tau\sigma^{-1}\right)(\alpha) = \alpha.$$

Thus, $\sigma\tau\sigma^{-1}$ fixes F, so $\sigma\tau\sigma^{-1} \in \mathrm{Gal}(K/F)$. Since τ was arbitrary, it follows that

$$\sigma \,\mathrm{Gal}(K/F)\sigma^{-1} \subseteq \mathrm{Gal}(K/F).$$

If we apply the same argument using σ^{-1} instead of σ, we get

$$\sigma^{-1}\operatorname{Gal}(K/F)\sigma \subseteq \operatorname{Gal}(K/F)$$
$$\Rightarrow \operatorname{Gal}(K/F) \subseteq \sigma \operatorname{Gal}(K/F)\sigma^{-1}.$$

Hence,

$$\sigma \operatorname{Gal}(K/F)\sigma^{-1} = \operatorname{Gal}(K/F)$$

which implies that $\sigma \in N$. Therefore $H \subseteq N$.
($N \subseteq H$) Let $\sigma \in N$ and by way of contradiction, suppose that $\sigma \notin H$. Then $\sigma(F) \neq F$, so there exists an element $\alpha \in F$ such that $\sigma(\alpha) \neq \alpha$. Hence there exists a nontrivial $\tau \in \operatorname{Gal}(K/F)$ such that $\tau(\sigma(\alpha)) \neq \sigma(\alpha)$ since $\operatorname{Gal}(K/F)$ only fixes F. Consider the conjugate

$$\sigma^{-1}\tau\sigma \in \operatorname{Gal}(K/F) \quad \because \sigma \in N$$

Thus this conjugate must fix F. Hence, for every $x \in F$,

$$\left(\sigma^{-1}\tau\sigma\right)(x) = x.$$

Applying σ on the left yields

$$(\tau\sigma)(x) = \sigma(x)$$

which contradicts the choice of $\tau \in \operatorname{Gal}(K/F)$. Therefore, we must have $\sigma \in H$. Hence $N \subseteq H$.
Thus $H = N$ as desired.

□

Cyclotomic Fields

Exercise 0.10. (Exercise 17)

a) Let k be a field of characteristic $\nmid 2n$, for some odd integer $n \geq 1$, and let ζ be a primitive n-th root of unity, in k. Show that k also contains a primitive $2n$-th root of unity.

b) Let k be a finite extension of the rationals. Show that there is only a finite number of roots of unity in k.

Proof. a) By assumption, $\operatorname{char}(k) \neq 2$, so $1 \neq -1$, and hence $-1 \in k$ has multiplicative order 2. Let $\zeta \in k$ be a primitive n-th root of unity with $n \geq 1$ odd. Consider the element $(-1)\zeta \in k^\times$. Suppose $((-1)\zeta)^m = 1$ for some positive integer m. Then

$$(-1)^m\zeta^m = 1 \implies \zeta^m = (-1)^m.$$

Since $(-1)^m \in \{1, -1\}$, we consider the two possibilities. If $\zeta^m = -1$, then

$$(\zeta^m)^2 = \zeta^{2m} = 1,$$

so $n \mid 2m$. But n is odd, so $n \mid m$, which would imply $\zeta^m = 1$, a contradiction. Hence $\zeta^m = 1$, so $n \mid m$. Then $(-1)^m = \zeta^m = 1$, so $2 \mid m$. Since $\gcd(2, n) = 1$, it follows that $2n \mid m$. Therefore, $(-1)\zeta$ has order exactly $2n$, and is a primitive $2n$-th root of unity in k.

b) ???

□

Exercise 0.11. (Exercise 19)
Let ζ be a primitive n-th root of unity. Let $K = \mathbb{Q}(\zeta)$.

a) If $n = p^r$ ($r \geq 1$) is a prime power, show that $N_{K/\mathbb{Q}}(1-\zeta) = p$.

b) If n is composite (divisible by at least two primes) then $N_{K/\mathbb{Q}}(1-\zeta) = 1$.

Proof. Immediately, observe that $\mathbb{Q}(\zeta)/\mathbb{Q}$ is Galois and, by Theorem 3.1, $\operatorname{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.

a) The irreducible polynomial of ζ over $\mathbb{Q}$ is

$$\operatorname{Irr}(\zeta, \mathbb{Q}, X) = \Phi_n(X) = \Phi_{p^r}(X) = \Phi_p\left(X^{p^{r-1}}\right) = \prod_{(k,n)=1} \left(X - \zeta^k\right)$$

Then since $\mathbb{Q}(\zeta)/\mathbb{Q}$ is separable,

$$N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(1-\zeta) = \prod_{\sigma} \sigma(1-\zeta), \quad \sigma \in \operatorname{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$$

For any $\sigma \in \operatorname{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$, $\sigma(1-\zeta) = \sigma(1) - \sigma(\zeta) = 1 - \zeta^k$ for $(k,n) = 1$. Hence,

$$N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(1-\zeta) = \prod_{(k,n)=1} \left(1 - \zeta^k\right) = \Phi_n(1)$$

Thus,

$$\Phi_n(1) = \Phi_{p^r}(1) = \Phi_p\left(1^{p^{r-1}}\right) = 1^{p-1} + 1^{p-2} + \cdots + 1^2 + 1 + 1 = p.$$

Therefore,

$$N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(1-\zeta) = \Phi_n(1) = p.$$

b) Let $n = p_1^{r_1} \cdots p_s^{r_s}$ with $p_i \in \mathbb{Z}$ distinct primes and $r_1 \geq 1$. Then from part (a), we still have

$$\Phi_n(X) = \prod_{(k,n)=1} \left(X - \zeta^k\right)$$

and

$$N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(1-\zeta) = \prod_{\sigma} \sigma(1-\zeta) = \prod_{(k,n)=1} \left(1 - \zeta^k\right)$$

Hence $N_{\mathbb{Q}(\zeta)/\mathbb{Q}} = \Phi_n(1)$. We wish to show that $\Phi_n(1) = 1$. We know

$$\begin{aligned} X^n - 1 &= \prod_{d|n} \Phi_d(X) \\ &= (X-1) \prod_{\substack{d|n, \\ d>1}} \Phi_d(X) \end{aligned}$$

Hence ,

$$\frac{X^n - 1}{X - 1} = \prod_{\substack{d|n, \\ d>1}} \Phi_d(X)$$

Taking the limit as $X \to 1$,

$$\lim_{X \to 1} \frac{X^n - 1}{X - 1} = \lim_{X \to 1} \frac{\frac{d}{dx}\left[X^n - 1\right]}{\frac{d}{dx}\left[X - 1\right]} = \lim_{X \to 1} nX^{n-1} = n$$

Thus,

$$
\begin{aligned}
n &= \prod_{\substack{d|n,\\ d>1}} \Phi_d(1) \\
&= \left(\prod_{\substack{d|n,\\ 1<d<n}} \Phi_d(1) \right) \Phi_n(1)
\end{aligned}
$$

Then

$$\Phi_n(1) = \frac{n}{\prod\limits_{\substack{d|n,\\ 1<d<n}} \Phi_d(1)}$$

Thus, if d is a prime power, $\Phi_d(1) = p$ and if d is a product of distinct primes, $\Phi_d(1) = 1$. Hence,

$$\prod_{\substack{d|n,\\ 1<d<n}} \Phi_d(1) = p_1^{r_1} \cdots p_s^{r_s} = n$$

Thus,
$\Phi_n(1) = \frac{n}{n} = 1$. Therefore, $N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(1-\zeta) = \Phi_n(1) = 1$.

□

Exercise 0.12. (Exercise 21)

a) Let a be a non-zero integer, p a prime, n a positive integer, and $p \nmid n$. Prove that $p \mid \Phi_n(a)$ if and only if a has period n in $(\mathbb{Z}/p\mathbb{Z})^\times$.

b) Again assume $p \nmid n$. Prove that $p \mid \Phi_n(a)$ for some $a \in \mathbb{Z}$ if and only if $p \equiv 1 \mod n$. Deduce from this that there are infinitely many primes $\equiv 1 \mod n$, a special case of Dirichlet's theorem for the existence of primes in an arithmetic progression.

Proof. a) ($\Longrightarrow$) Assume that $p \mid \Phi_n(a)$. Thus $\Phi_n(a) \equiv 0 \mod p$. Consider the factorization

$$X^n - 1 = \prod_{d|n} \Phi_d(X).$$

Evaluating at a and reducing modulo p, and from our assumption that $\Phi_n(a) \equiv 0 \mod p$, we obtain

$$
\begin{aligned}
a^n - 1 &\equiv 0 \mod p \\
a^n &\equiv 1 \mod p
\end{aligned}
$$

Let r be the order of $a \mod p$. Hence $r \mid n$. We wish to show that $r = n$. By way of contradiction, suppose that $r < n$. Then since $a^r \equiv 1 \mod p$,

$$a^r - 1 = \prod_{d|r} \Phi_d(a) \equiv 0 \mod p.$$

Hence $p \mid \Phi_d(a)$ for some $d \mid r$. But since $r < n$, every such d is less than n. Thus, some cyclotomic polynomial of degree less than that of $\Phi_n(X)$ reduces to $0 \mod p$ (evaluated at a). This contradicts the assumption that $p \mid \Phi_n(a)$ and $p \nmid \Phi_d(a)$ for all $d < n$. Therefore since $r \not< n$ and $r \mid n$, we must have that $r = n$. Hence, a has order n in $(\mathbb{Z}/p\mathbb{Z})^\times$.

($\Longleftarrow$) Assume that a has order n modulo p. Hence, $a^n \equiv 1 \mod p$ and $a^d \not\equiv 1 \mod p$ for $d < n$. Then

$$a^n - 1 = \prod_{d|n} \Phi_d(a)$$

Reducing modulo p,

$$a^n - 1 = \prod_{d|n} \Phi_d(a) \equiv 0 \mod p.$$

Thus, since $a^d \not\equiv 1 \mod p$ for $d < n$, $\Phi_d(a) \not\equiv 0 \mod p$ which implies that we must have $\Phi_n(a) \equiv 0 \mod p$ and thus $p \mid \Phi_n(a)$.

b) By part (a),

$$\Phi_n(a) \equiv 0 \mod p \iff a^n \equiv 1 \mod p.$$

By Fermat's Little Theorem and Lagrange's Theorem, we have $a^{\varphi(p)} = a^{p-1} \equiv 1 \mod p$ for all a such that $(a,p) = 1$. Then $n \mid |(\mathbb{Z}/p\mathbb{Z})^\times| \Rightarrow n \mid \varphi(p) \Rightarrow n \mid p-1 \Rightarrow p \equiv 1 \mod n$.

We know deduce the special case of Dirichlet's theorem. By way of contradiction, assume that there are finitely many primes $p_1, \ldots, p_k$ such that $p_i \equiv 1 \mod n$. Set $P = \{p_1, \ldots, p_k\}$. For a sufficiently large $M \in \mathbb{N}$, we have

$$\Phi_n(Mp_1 \cdots p_k) > 1$$

such that $\Phi_n(Mp_1 \cdots p_k) \equiv 0 \mod p$ for some prime p. We see that $p \neq p_i$ for all $i = 1, \ldots, k$ since by definition

$$\Phi_n(Mnp_1 \cdots p_k) \equiv 1 \mod p_i.$$

Hence, we found a distinct $p \notin P$, implying that the set of primes dividing $\Phi_n(a)$ as a ranges over $\mathbb{Z}$ is infinite. Therefore, from part (a), we conclude that P is infinite yielding a special case of Dirichlet's theorem for the existence of primes in an arithmetic progression.

□

Exercise 0.13. (Exercise 29)

a) Let K be a cyclic extension of a field F, with Galois group G generated by σ. Assume that the characteristic is p, and that $[K\colon F] = p^{m-1}$ for some integer $m \geq 2$. Let β be an element of K such that $\mathrm{Tr}_F^K(\beta) = 1$. Show that there exists an element α in K such that

$$\sigma\alpha - \alpha = \beta^p - \beta.$$

b) Prove that the polynomial $X^p - X - \alpha$ is irreducible.

c) If θ is a root of this polynomial, prove that $F(\theta)$ is a Galois, cyclic extension of degree p^m of F, and that its Galois group is generated by an extension σ^* of σ such that

$$\sigma^*(\theta) = \theta + \beta.$$

Proof. Assume that $G = \mathrm{Gal}(K/F) = \langle\sigma\rangle$.

a) Let $\beta \in K$ such that $\mathrm{Tr}_F^K(\beta) = 1$. By definition, $\sum_{i=1}^{p^{m-1}} \sigma^i(\beta) = 1$. Then

$$\begin{aligned}\mathrm{Tr}_F^K(\beta^p) &= \sum_{i=1}^{p^{m-1}} \sigma^i(\beta^p) \\ &= \sum_{i=1}^{p^{m-1}} \sigma^i(\beta)^p \\ &= \left(\sum_{i=1}^{p^{m-1}} \sigma^i(\beta)\right)^p \qquad \because \operatorname{char} F = p \\ &= \left(\mathrm{Tr}_F^K(\beta)\right)^p.\end{aligned}$$

Therefore,

$$\begin{aligned}\mathrm{Tr}_F^K(\beta^p - \beta) &= \mathrm{Tr}_F^K(\beta^p) - \mathrm{Tr}_F^K(\beta) \\ &= \left(\mathrm{Tr}_F^K(\beta)\right)^p - \mathrm{Tr}_F^K(\beta) \\ &= 1^p - 1 = 0.\end{aligned}$$

Therefore, by Theorem 6.3, there exists some $\alpha \in K$ such that $\beta^p - \beta = \alpha - \sigma\alpha$. Note that we can turn this into what the problem asks by simply substituting $-\alpha$ for α instead.

b) By Theorem 6.4(ii), either $X^p - X - \alpha \in K[X]$ has a root in K, and thus all roots are in K, or it is irreducible. We show that a root θ is not in K so that $X^p - X - \alpha$ is irreducible. By way of contradiction, assume that $\theta \in K$ is a root of $X^p - X - \alpha$, that is, $\theta^p - \theta = \alpha$. Applying σ,

$$\sigma\theta^p - \sigma\theta = \sigma\alpha.$$

Subtracting α from both sides yields

$$\begin{aligned}\sigma\theta^p - \sigma\theta - \alpha &= \sigma\alpha - \alpha \\ \sigma\theta^p - \sigma\theta - (\theta^p - \theta) &= \sigma\alpha - \alpha \\ (\sigma\theta - \theta)^p - (\sigma\theta - \theta) &= \sigma\alpha - \alpha = \beta^p - \beta.\end{aligned}$$

Thus,

$$\begin{aligned}(\sigma\theta - \theta - \beta)^p - (\sigma\theta - \theta - \beta) &= 0 \\ (\sigma\theta - \theta - \beta)^p &= \sigma\theta - \theta - \beta.\end{aligned}$$

Hence, $\sigma\theta - \theta - \beta \in \mathbb{F}_p \subseteq F \subseteq K$ (by $x^p = x$ for all $x \in \mathbb{F}_p$), i.e. $\sigma\theta - \theta - \beta \in F$. Since $F = K^{\mathrm{Gal}(K/F)}$ and $[K \colon F] = p^{m-1}$, $\mathrm{Tr}_F^K(\gamma) = p^{m-1}\gamma = 0$ for all $\gamma \in F$. It follows that

$$\begin{aligned}\mathrm{Tr}_F^K(\sigma\theta - \theta - \beta) &= 0 \\ \mathrm{Tr}_F^K(\sigma\theta - \theta) &= \mathrm{Tr}_F^K(\beta) \\ 0 &= 1,\end{aligned}$$

an absurdity. Thus, $\theta \notin K$ so by Theorem 6.4(ii), $X^p - X - \alpha \in K[X]$ is irreducible.

c) I believe the statement is best understood by considering $K(\theta)$. By part (b), we know that $X^p - X - \alpha \in K[X]$ is irreducible. Let θ be a root of this polynomial in some algebraic closure K^a and consider the extension $K(\theta)$. Since $\frac{d}{dX}[X^p - X - \alpha] = -1 \not\equiv 0 \mod p$, $K(\theta)/K$ is separable. Observe that $\theta + k$ is also a root for all $k \in \mathbb{F}_p \subseteq K(\theta)$ and thus $K(\theta)$ contains all of the roots of $X^p - X - \alpha$ which implies that $K(\theta)/K$ is normal and hence Galois. Therefore, $K(\theta)/F$ is Galois since each step in the tower $F \subseteq K \subseteq K(\theta)$ is Galois. By the tower law,

$$[K(\theta)\colon F] = [K(\theta)\colon K][K\colon F] = p \cdot p^{m-1} = p^m.$$

Define $\sigma^*\colon K(\theta) \to K(\theta)$ by $\theta \mapsto \theta + \beta$. Observe that $\sigma^*\big|_K = \sigma$. Furthermore, since $|\sigma| = p^{m-1}$, consider

$$\begin{aligned}(\sigma^*)^{p^{m-1}}(\theta) &= \theta + \sum_{i=1}^{p^{m-1}} \sigma^i(\beta) \\ &= \theta + \mathrm{Tr}_F^K(\beta) \\ &= \theta + 1.\end{aligned}$$

Thus, it follows that $\left|(\sigma^*)^{p^{m-1}}\right| = p$ and therefore $|\sigma^*| = p^m = [K(\theta)\colon F] = |\operatorname{Gal}(K(\theta)/F)|$. Hence,

$$\operatorname{Gal}(K(\theta)/F) = \langle \sigma^* \rangle \cong \mathbb{Z}/p^m\mathbb{Z},$$

which shows that $K(\theta)/F$ is cyclic as desired.

□

Exercise 0.14. (Exercise 30)
Let A be an abelian group and let G be a finite cyclic group operating on A [by means of a homomorphism $G \to \mathrm{Aut}(A)$]. Let σ be a generator of G. We define the trace $\mathrm{Tr}_G = \mathrm{Tr}$ on A by $\mathrm{Tr}(x) = \sum_{\tau \in G} \tau x$. Let A_{Tr} denote the kernel of the trace, and let $(1-\sigma)A$ denote the subgroup of A consisting of all elements of type $y - \sigma y$. Show that $H^1(G, A) \cong A_{\mathrm{Tr}}/(1-\sigma)A$.

Proof. Let $|G| = n$. Define a homomorphism

$$\zeta\colon Z^1(G, A) \to A$$

by $\alpha \mapsto \alpha_\sigma$ since every element in G is determined by σ. We first show that ζ is injective. Let $\alpha \in \ker\zeta$. Then $\zeta(\alpha) = \alpha_\sigma = 0$. By definition of $\alpha \in Z^1(G, A)$, $\alpha_\sigma + \sigma^k\alpha_\sigma = \alpha_{\sigma^{k+1}}$ for all $0 \le k \le n-1$. Hence,

$$\begin{aligned}&(k=0) \quad \alpha_\sigma = 0 \\ &(k=1) \quad \alpha_{\sigma^2} = \alpha_\sigma + \sigma\alpha_\sigma = 0 + \sigma(0) = 0 \\ &(k=2) \quad \alpha_{\sigma^3} = \alpha_\sigma + \sigma^2\alpha_\sigma = 0 + \sigma^2(0) = 0 \\ &\cdots\cdots \quad \cdots\cdots \quad \cdots\cdots \\ &(k=n-1) \quad \alpha_{\sigma^n} = \alpha_\sigma + \sigma^{n-1}\alpha_\sigma = 0 + \sigma^{n-1}(0) = 0.\end{aligned}$$

Therefore, $\alpha_{\sigma^k} = 0$ for all $0 \le k \le n-1$, i.e. for every element of G, it follows that $\alpha = 0 \in Z^1(G, A)$ is the zero map. Thus, $\ker\zeta = \{0\}$ so ζ is injective.
Next, we will show that ζ maps $Z^1(G, A)$ onto A_{Tr}. Let $\alpha \in Z^1(G, A)$. Then $\zeta(\alpha) = \alpha_\sigma$. Applying the trace, we obtain

$$\begin{aligned}\mathrm{Tr}(\zeta(\alpha)) = \mathrm{Tr}(\alpha_\sigma) &= \sum_{k=0}^{n-1} \sigma^k \alpha_\sigma \\ &= \alpha_\sigma + \sigma\alpha_\sigma + \sigma^2\alpha_\sigma + \cdots + \sigma^{n-1}\alpha_\sigma.\end{aligned}$$

By repeated use of the cocycle relation, we have

$$\begin{aligned}
\alpha_{\sigma^n} &= \alpha_\sigma + \sigma\alpha_{\sigma^{n-1}} \\
&= \alpha + \sigma(\alpha_\sigma + \sigma\alpha_{\sigma^{n-2}}) \\
\cdots\cdots & \quad \cdots\cdots \\
&= \alpha_\sigma + \sigma\alpha_\sigma + \sigma^2\alpha_\sigma + \cdots + \sigma^{n-1}\alpha_\sigma.
\end{aligned}$$

But $\alpha_{\sigma^n} = \alpha_e = 0$ (identity maps to identity), so we have that $\mathrm{Tr}(\zeta(\alpha)) = \mathrm{Tr}(\alpha_\sigma) = \alpha_{\sigma^n} = 0$. Hence,

$$\zeta(Z^1(G,A)) \subseteq A_{\mathrm{Tr}}.$$

Now let $a \in A_{\mathrm{Tr}}$, that is, $a + \sigma a + \cdots + \sigma^{n-1}a = 0$. Define $\alpha\colon G \to A$ by

$$\alpha_{\sigma^k} = \sum_{i=0}^{k-1} \sigma^i(a)$$

which is well-defined, satisfies the cocycle relation, and $\alpha_\sigma = a$. Hence, $\alpha \in Z^1(G,A)$. Thus,

$$A_{\mathrm{Tr}} \subseteq \zeta(Z^1(G,A)),$$

so $A_{\mathrm{Tr}} = \zeta(Z^1(G,A)) \subseteq A$. Therefore, injectivity is preserved if we restrict the codomain of ζ to its image. Then we have an isomorphism

$$Z^1(G,A) \cong A_{\mathrm{Tr}}.$$

We now show a correspondence between $B^1(G,A)$ and $(1-\sigma)A$. Let $\alpha \in B^1(G,A)$. Then

$$\begin{aligned}
\zeta(\alpha) = \alpha_\sigma &= \sigma b - b \\
&= (\sigma - 1)b \\
&= -(1-\sigma)b \in (1-\sigma)A
\end{aligned}$$

Hence, $B^1(G,A) \triangleleft Z^1(G,A)$ corresponds to $(1-\sigma)A \triangleleft A_{\mathrm{Tr}}$ (trivially $(1-\sigma)A \leqslant A_{\mathrm{Tr}}$). Thus, combining $B^1(G,A) = (1-\sigma)A$ with the previous isomorphism

$$\begin{aligned}
Z^1(G,A)/B^1(G,A) &\cong A_{\mathrm{Tr}}/(1-\sigma)A \\
H^1(G,A) &\cong A_{\mathrm{Tr}}/(1-\sigma)A.
\end{aligned}$$

□

Exercise 0.15. (Exercise 32)
Let E be a finite separable extension of k, of degree n. Let $W = (w_1, \ldots, w_n)$ be elements of E. Let $\sigma_1, \ldots, \sigma_n$ be the distinct embeddings of E in k^a over k. Define the **discriminant** of W to be

$$D_{E/k}(W) = \det(\sigma_i w_j)^2.$$

Prove:

a) If $V = (v_1, \ldots, v_n)$ is another set of elements of E and $C = (c_{ij})$ is a matrix of elements of k such that $w_i = \sum c_{ij} v_j$, then

$$D_{E/k}(W) = \det(C)^2 D_{E/k}(V).$$

b) The discriminant is an element of k.

c) Let $E = k(\alpha)$ and let $f(X) = \mathrm{Irr}(\alpha, k, X)$. Let $\alpha_1, \dots, \alpha_n$ be the roots of f and say $\alpha = \alpha_1$. Then

$$f'(\alpha) = \prod_{j=2}^{n} (\alpha - \alpha_j).$$

Show that

$$D_{E/k}(1, \alpha, \dots, \alpha^{n-1}) = (-1)^{n(n-1)/2} N_k^E(f'(\alpha)).$$

d) Let the notation be as in (a). Show that $\det(\mathrm{Tr}(w_i w_j)) = (\det(\sigma_i w_j))^2$. [*Hint*: Let A be the matrix $(\sigma_i w_j)$. Show that ${}^tAA$ is the matrix $(\mathrm{Tr}(w_i w_j))$.]

Proof. a) By assumption,

$$W = CV.$$

Let $w_j = \sum_{k=1}^{n} c_{jk} v_k$. Then

$$\begin{aligned}
D_{E/k}(W) = D_{E/k}(CV) &= \det(\sigma_i w_j)^2 \\
&= \det\left(\sigma_i \left(\sum c_{jk} v_k\right)\right)^2 \\
&= \det(C_{(i)} \cdot \sigma_i V)^2 \\
&= \det((\sigma_i v_j)^t C)^2 \\
&= \det(\sigma_i v_j)^2 \det(C)^2 \\
&= D_{E/k}(V) \det(C)^2 = \det(C)^2 D_{E/k}(V).
\end{aligned}$$

b) Let $\tau \in \mathrm{Gal}(k^a/k)$ (or use some minimal extension of E contained in k^a that is also normal). We wish to show that $\tau D_{E/k}(W) = D_{E/k}(W)$ since k is the fixed field of the Galois group. Thus,

$$\tau D_{E/k}(W) = \tau \left(\det(\sigma_i w_j)\right)^2 = \det((\tau \circ \sigma_i) w_j)^2.$$

Note that we can pass τ into the determinant, a multilinear map, by Ch. XIII Corollary 4.7. Then $\tau \circ \sigma_i \colon E \hookrightarrow k^a$ is again an embedding and since $\sigma_1, \dots, \sigma_n$ are the distinct embeddings, $\tau \circ \sigma_i$ must be one of $\sigma_1, \dots, \sigma_n$. Thus, τ permutes σ_i and furthermore permutes the rows of the matrix $(\sigma_i w_j)$. Since τ permutes σ_i, there exists some $\pi_\tau \in S_n$ (determined by τ) such that $\tau \circ \sigma_i = \sigma_{\pi_\tau(i)}$ for all $i = 1, \dots, n$. Thus,

$$\begin{aligned}
\det((\tau \circ \sigma_i) w_j)^2 &= (\varepsilon(\pi_\tau) \det(\sigma_i w_j))^2 \\
&= \varepsilon(\pi_\tau)^2 \det(\sigma_i w_j)^2 \\
&= \det(\sigma_i w_j)^2 \\
&= D_{E/k}(W)
\end{aligned}$$

where $\varepsilon(\pi_\tau)$is the sign of the permutation $\pi_\tau \in S_n$, so $\varepsilon(\pi_\tau)^2 = 1$. Thus, since $\tau \in \mathrm{Gal}(k^a/k)$ was arbitrary, $D_{E/k}(W)$ is fixed by every automorphism and therefore $D_{E/k}(W) \in k$.

c) Let $w_j = \alpha^j$ for all $j = 0, 1, \dots, n-1$. Thus, $\sigma_i(\alpha) = \alpha_i$. Then, our matrix $(\sigma_i w_j)$ is given by

$$
\begin{aligned}
(\sigma_i w_j) &= \begin{pmatrix} \sigma_1 w_0 & \cdots & \sigma_1 w_{n-1} \\ \vdots & \ddots & \vdots \\ \sigma_n w_0 & \cdots & \sigma_n w_{n-1} \end{pmatrix} \\
&= \begin{pmatrix} \sigma_1(1) & \cdots & \sigma_1(\alpha^{n-1}) \\ \vdots & \ddots & \vdots \\ \sigma_n(1) & \cdots & \sigma_n(\alpha^{n-1}) \end{pmatrix} \\
&= \begin{pmatrix} 1 & \alpha_1 & \alpha_1^2 & \cdots & \alpha_1^{n-1} \\ 1 & \alpha_2 & \alpha_2^2 & \cdots & \alpha_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \cdots & \alpha_n^{n-1} \end{pmatrix}.
\end{aligned}
$$

Hence, $(\sigma_i w_j)$ is a Vandermonde matrix, so $\det(\sigma_i w_j)^2 = \left(\prod_{i<j}(\alpha_j - \alpha_i)\right)^2 = \prod_{i<j}(\alpha_i - \alpha_j)^2$. The norm of $f'(\alpha)$ is given by

$$
\begin{aligned}
N_k^E(f'(\alpha)) &= \prod_{i=1}^{n} \sigma_i f'(\alpha) \\
&= \prod_{i=1}^{n} f'(\sigma_i \alpha) \\
&= \prod_{i=1}^{n} f'(\alpha_i).
\end{aligned}
$$

Observe that $f'(\alpha_i) = \prod_{j \neq i}(\alpha_i - \alpha_j)$. Thus,

$$
\begin{aligned}
N_k^E(f'(\alpha)) &= \prod_{i=1}^{n} \prod_{j \neq i}(\alpha_i - \alpha_j) \\
&= \prod_{i<j}(\alpha_i - \alpha_j)(\alpha_j - \alpha_i) \\
&= \prod_{i<j}(-1)(\alpha_i - \alpha_j)^2.
\end{aligned}
$$

There are precisely $\binom{n}{2} = \frac{n(n-1)}{2}$ pairs of (i, j), so we obtain

$$
\begin{aligned}
N_k^E(f'(\alpha)) &= (-1)^{\binom{n}{2}} \prod_{i<j}(\alpha_i - \alpha_j)^2 \\
&= (-1)^{n(n-1)/2} \prod_{i<j}(\alpha_i - \alpha_j)^2 \\
&= (-1)^{n(n-1)/2} D_{E/k}(W).
\end{aligned}
$$

Thus, dividing by $(-1)^{n(n-1)/2}$, we obtain

$$
D_{E/k}(W) = D_{E/k}(1, \alpha, \ldots, \alpha^{n-1}) = (-1)^{n(n-1)/2} N_k^E(f'(\alpha)).
$$

d) We follow the hint. Let $A = (\sigma_i w_j)$, that is,

$$A = \begin{pmatrix} \sigma_1 w_i & \cdots & \sigma_1 w_n \\ \vdots & \ddots & \vdots \\ \sigma_n w_1 & \cdots & \sigma_n w_n \end{pmatrix}.$$

Then the transpose is given by

$$^tA = \begin{pmatrix} \sigma_1 w_1 & \cdots & \sigma_n w_1 \\ \vdots & \ddots & \vdots \\ \sigma_1 w_n & \cdots & \sigma_n w_n \end{pmatrix}.$$

Taking their product,

$$\begin{aligned} ^tAA &= \begin{pmatrix} \sum_{i=1}^{n} \sigma_i(w_1^2) & \cdots & \sum_{i=1}^{k} \sigma_i(w_1 w_n) \\ \vdots & \ddots & \vdots \\ \sum_{i=1}^{n} \sigma_i(w_n w_1) & \cdots & \sum_{i=1}^{n} \sigma_i(w_n^2) \end{pmatrix} \\ &= \begin{pmatrix} \mathrm{Tr}(w_1^2) & \cdots & \mathrm{Tr}(w_1 w_n) \\ \vdots & \ddots & \vdots \\ \mathrm{Tr}(w_n w_1) & \cdots & \mathrm{Tr}(w_n^2) \end{pmatrix} \\ &= (\mathrm{Tr}(w_i w_j)). \end{aligned}$$

Therefore,

$$\begin{aligned} \det(\mathrm{Tr}(w_i w_j)) &= \det(^tAA) \\ &= \det(^tA)\det(A) \\ &= \det(A)^2 \\ &= \det(\sigma_i w_j)^2 \\ &= D_{E/k}(W). \end{aligned}$$

□